MIL-P-14999 (MU)
28 April 1969
SEE SUPERSESSION
DATA - 6.5

MILITARY SPECIFICATION

POWDER, MOLDING COMPOUND EXPLOSIVE (PBX)

1. SCOPE

1.1 This Military Specification covers two types of molding powder which are used to press pellets for use in ammunition.

1.2 Classification.-The PBX shall be of the following types:

Type I	Consisting of RDX, Polystyrene, and Dioctylphthalate
Type II	Consisting of RDX and Plastic-Plasticizer

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

LLL-R-626	-	Rosin, Gum, Rosin, Wood, and Rosin, Tall Oil
RR-S-366	-	Sieves; Standard, Testing

MILITARY

MIL-R-398	-	RDX
MIL-D-13796	-	Dioctylphthalate
MIL-P-55025	-	Polystyrene, Unmodified (For Use as a Binder in Explosives)

FSC: 1376

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

7548644	Box, Packing for High Explosives Assembly; Details, Packing and Marking
7548645	Carton, Packing, Reusable-Collapsible, for High Explosives
7548646	Carton, Packing, Reusable-Collapsible, Alternative, for High Explosives, Assembly, Details, Packing and Marking

2.2 Other publication.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS, Title 49: Transportation; Parts 171-179

The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. Orders for the above publication should cite: "49 CFR 171-179 (latest revision)."

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material

3.1.1 Type I.-The molding powder (PBX) shall be a mixture of RDX, coated and surrounded by a homogeneous mixture of polystyrene and dioctylphthalate. The constituent materials used in the manufacture of molding powder shall comply with the following requirements:

Constituent Material:	Conforming to:
RDX	Specification MIL-R-398, Type B
Polystyrene (Unmodified)	Specification MIL-P-55025
Dioctylphthalate	Specification MIL-D-13796

3.1.2 Type II.-The molding powder shall be a mixture of RDX and plastic plasticizer. The constituent materials used in the manufacture of molding powder shall comply with the following requirements:

Constituent Material:	Conforming to:
RDX	Specification MIL-R-398, Type B
Plastic-Plasticizer	
Polystyrene (Unmodified)	Specification MIL-P-55025
Dioctylphthalate	Specification MIL-D-13796
Gum Rosin	Specification LLL-E-626 (see 6.2)

3.2 Composition.-The composition of molding powder shall be as shown in Table I, when determined as specified in the applicable paragraph, 4.3.1 to 4.3.4.

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TABLE I
Composition

Constituent	Type I - Percent	Type II - Percent
RDX	90.0 ± 0.5	90.0 ± 0.5
Polystyrene	8.5 ± 0.3	------
Dioctylphthalate	1.5 ± 0.3	------
Plastic-plasticizer	-------	10.0 ± 0.5*

*The specified percentage of plastic-plasticizer shall consist of a mixture of 91 ± 1 percent polystyrene, 5.0 ± 1 percent dioctylphthalate and 4.0 ± 1 percent gum rosin.

3.3 Foreign matter.-The molding powder shall be free of inclusions and particles when determined as specified in 4.3.5.

3.4 Color (applicable only to Type II).-When specified by the procuring agency, the molding powder shall be colored pink (see 6.3). The color produced shall match a sample color furnished by the procuring agency when determined as specified in 4.3.6.

3.5 Moisture.-The moisture content shall be 0.3 percent maximum (max.) when determined as specified in 4.3.7.

3.6 Granulation.-The granulation of the molding powder shall be as indicated in Table II, when determined as specified in 4.3.8.

TABLE II
Granulation

U. S. Standard Sieve No.	Type I Percent, Minimum	Type II Percent, Minimum
18	Through 100	Through 100
30	Through 95	Through 95
60	Retained 90	Retained 90
80	Retained 98	Retained 98
100	Retained 99.9	Retained 99.5

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Drawing and specification number and date, together with an identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous quantity of PBX from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample

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selected shall represent only that quantity of PBX from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more batches of PBX produced by one manufacturer, in accordance with the same specification or same revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of PBX subjected to the same unit chemical or physical process intended to make the final produce homogeneous. Unless otherwise approved by the contracting officer, the inspection lot size of explosive deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract perior nor more than the largest quantity contractually scheduled for delivery during any month of the contract period.

4.2.2 Examination.-Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans and procedures for major and minor defects shall be in accordance with Standard MIL-STD-105, except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristic listed using an AQL of 0.65 percent for each major defect, except where 100 percent inspection is specified.

4.2.2.1 Wooden box or fiberboard carton, prior to closing (see drawing 7548644, 7548645, and 7548646).

Categories	Defects	Method of Inspection	Code No. (See 6.4)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Liner pierced or torn	Visual	01001
102.	Liner improperly closed	Visual	01002
103.	Foreign matter	Visual	01003
Minor:	AQL 0.65 percent		
201.	Type of liner incorrect	Visual	01004

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Minor:	AQL 0.65 percent		
201.	Type of liner incorrect.....	Visual	01004

4.2.2.2 Sealed wooden box (see drawing 7548645).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Box damaged.....	Visual	02001
102.	Top improperly assembled.....	Visual	02002
103.	Strapping broken or loose.....	Visual-Manual	02003
Minor:	AQL 1.50 percent		
201.	Nail protruding.....	Visual	02004
202.	Marking misleading or unidentifiable.....	Visual	02005
203.	Strapping improperly assembled...	Visual-Manual	02006

4.2.2.3 Sealed fiberboard carton (see drawing 7548645 and 7548646).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Assembly torn or pierced.....	Visual	03001
102.	Strapping broken or loose.....	Visual-Manual	03002
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable.....	Visual	03003

4.2.3.1 Sampling by lot.-A random sample of 500 grams of the PBX shall be prepared by selecting equal quantities of the PBX from at least 10 randomly selected containers. The individual primary samples shall then be added to a sample container. (one sample shall be added on top of the other sample). When lots are comprised of 10 containers or less, each container shall be sampled. The prepared sample shall be used for tests described in 4.3. If the sample fails to comply with any of the requirements (see 3.1 to 3.6) specified, the lot shall be rejected.

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4.3 Test methods and procedures

4.3.1 RDX - Code No. 04001.

4.3.1.1 Reagents

Amalgamated Zinc: Dissolve about 10 to 15 g of mercury chloride in a liter of 15 - 18% HCl. Pour the solution into a beaker containing 1 - 2 lbs of mossy zinc and allow to react for about 20 minutes with the zinc completely immersed. Decant the solution and wash the amalgamated zinc several times, using a total of about a liter of distilled water. Store the amalgamated zinc under water in an inert atmosphere. It can be reused many times.

Chromous Chloride (0.2N): Dissolve 110 g of $CrCl_3.6H_2O$ in about 500 ml of distilled water contained in a 2-liter volumetric flask. Add 30 ml of concentrated HCl and dilute to the mark with distilled water. After mixing, pour the solution into a 4-liter bottle containing the amalgamated zinc. Allow the mixture to react overnight under an inert atmosphere and maintain this atmosphere at all times. A bright blue color indicates complete reduction of chromic (green) to chromous (blue) ions. Store and standardize the solution as described in MIL-STD-286, Method No. 601.1, (Propellants: Sampling, Inspection and Testing) but omit the correction for iron.

Ferric Alum (0.15N): Prepare and standardize as directed in MIL-STD-286, Method 603.1, (Propellants: Sampling, Inspection and Testing).

Phenosafranin Indicator: Dissolve 2 grams of phenosafranin in 100 ml of distilled water.

Glacial Acetic Acid: ACS Grade

Inert Gas (Oxygen-Free): CO_2 or N_2

Concentrated Hydrochloric Acid, 36%: ACS Grade

Hydrochloric Acid, 15%: Add 400 ml of the concentrated HCl to 600 ml of water.

4.3.1.2 Procedure.-Transfer approximately 0.5g of the sample accurately weighed to 0.001g, to a 250 ml volumetric flask. Add about 200 ml of acetic acid and heat on a hot plate until the RDX is dissolved. Cool the flask and contents to room temperature and dilute to volume with acetic acid. Pipet a 25 ml aliquot into a 500 ml titration flask. Displace the

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air with a stream in inert gas and maintain the flow of this gas throughout the determination. Add 25 ml of concentrated HCl and stir the contents for 10 minutes. Add exactly 100 ml of 0.2N chromous chloride solution to the titration flask and stir again for 10 minutes. Add 15 drops of phenosafranin indicator solution and titrate with 0.5N ferric alum solution until the end point, which is a sharp change from green to deep red.

Run a blank determination following the same procedure but omitting the sample.

Calculate the percentage of RDX as follows:

$$RDX\% = \frac{12.34\ (V_1 - V_2)\ N}{W}$$

Where:

V_1 = Volume of ferric ammonium sulfate used to titrate the blank, ml.
V_2 = Volume of ferric ammonium sulfate used to titrate the excess chromous chloride, ml.
W = Weight of sample, gm. (Moisture-free).
N = Normality of the ferric ammonium sulfate solution.

4.3.2 Plastic plasticizer, Code No. 05001 (NOTE: Use this calculation for Type II only).

Calculate the percentage of Plastic-plasticizer as follows:

Plastic-plasticizer, % = 100 - (percentage of RDX found)

4.3.3 Dioctylphthalate, Code No. 06001.-Transfer into a 250 ml beaker an accurately weighed portion of approximately 2 grams of a fresh sample. Add 100 ml of isopropanol saturated with RDX to the beaker, cover the beaker, and place on a steam bath for 1/2 hour. Remove beaker with sample and quantitatively transfer the contents to a tared medium porosity glass filtering crucible on a vacuum filtrator assembly. Use vacuum to aid in filtering. Wash the contents of the crucible with three-20 ml portions of isopropanol saturated with RDX. Air dry the crucible and contents. Dry the crucible in an oven maintained at 90 plus or minus 2 degrees C for one hour, then cool and weigh the crucible. Calculate the percentage of dioctylphthalate as follows:

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$$\text{Dioctylphthalate, \%} = \frac{(B - A)\ 100}{W}$$

Where:

B = Weight of crucible plus sample
A = Weight of crucible and residue after extraction
W = Weight of sample, gms. (Moisture Free).

4.3.4 Polystyrene, Code No. 07001 (NOTE: This calculation shall be used for Type I).

Calculate the percentage of polystyrene as follows:

Polystyrene, % = 100-(percent RDX found; percent dioctylphthalate found).

4.3.5 Foreign matter, Code No. 08001.-The molding powder shall be visually inspected for metallic inclusion and wooden particles as well as other foreign matter. (NOTE: This determination shall be performed after all the other chemical determinations have been completed. Five hundred grams of the remaining sample shall be spread on a sheet of paper (approximately 3 feet by 2 feet) and with the eye examined for presence of inclusions and particles).

4.3.6 Determination of color, Code No. 09001.-(Applicable to Type II only).-The color shall be determined visually.

4.3.7 Determination of moisture, Code No. 10001.-The determination shall be as follows: Transfer an accurately weighed portion of approximately 5 gram (gm) of the sample into a previously dried, aluminum dish (approximately 60 millimiters (mm) in diameter and 30 mm deep. Heat the sample in a vacuum oven (pressure not greater than 8 centimeters mercury) at a temperature maintained at 55 plus or minus 2 degrees C for 4 hours. Calculate the percent moisture from the loss in weight as follows:

$$\text{Percent moisture} = \frac{A}{B} \times 100$$

Where:

A = Loss in weight of the sample, in gm.
B = Weight of sample, in gm.

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4.3.8 Granulation, Code No. 11001.-A 100 plus or minus 0.1 gm portion of the sample shall be placed on the specified sieve or nest of sieves, complying with Specification RR-S-366, properly superimposed and assembled with a bottom pan. The sieve or nest of sieves shall be covered and shaken for three minutes by means of a mechanically operated sieve shaker, which shall impart to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. The portions retained or passed by the various sieves shall be weighed and the results calculated to a cumulative percentage basis.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.1)

5.1.1 Level A.-The explosive compound shall be packed and marked in accordance with drawing 7548644.

5.1.2 Level C.-The explosive compound shall be packed and marked in accordance with drawings 7548645, 9548646 or as applicable.

5.2 Packing, Level C.-The PBX shall be packed in uniform quantities in conformance with the manufacturer's commercial practice as well as in accordance with Interstate Commerce Commision Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles (49 CFR 171-190). The PBX should be packed to provide protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers used shall comply with Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation. Containers shall be of uniform size, shape and material.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Levels of protection (see 5.1.1 and 5.1.2).
c. Type of PBX required.

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6.2 Taylor Lowenstein Company, Mobile, Alabama, Grade WW, Commercial Walter White Rosin has been found a satisfactory material for use in PBX. Any other material proposed for this use should be submitted to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey, 07801, for determination of compatability with the explosives.

6.3 Approximately 2.1 pounds of Sherman Williams Naphthol Red. No. 10397 per thousand (pounds) mix has been found satisfactory for meeting the requirements of 3.4. Any other material proposed for this use should be submitted to: Commanding Officer, Picatinny Arsenal, SMUPA-ND2, Dover, New Jersey, 07801, for determination of compatability with the explosives.

6.4 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 MIL-P-14999 (MU), supersedes PA-PD-711C dated 23 August 1966.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1376-A022

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